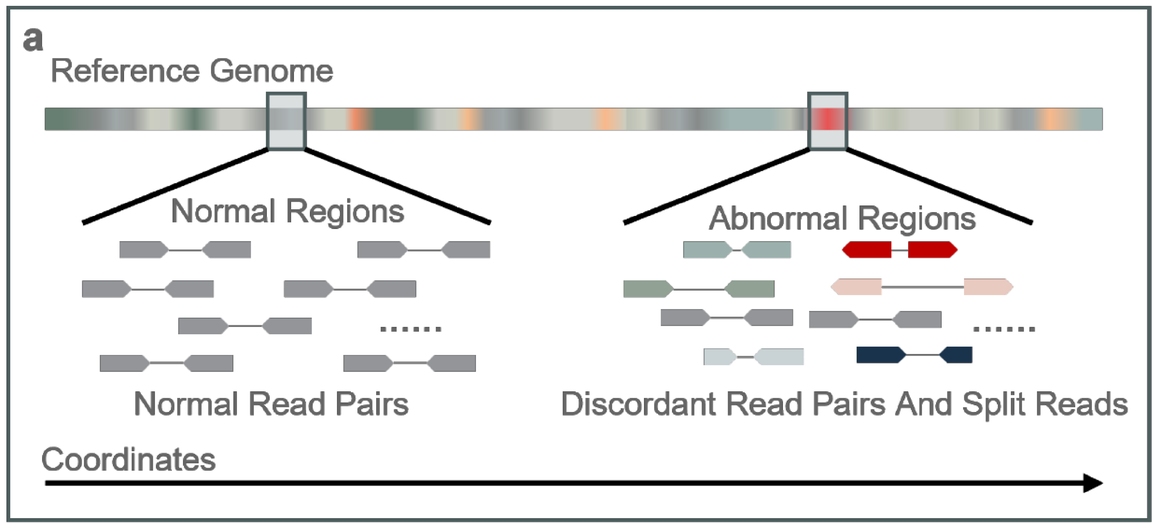
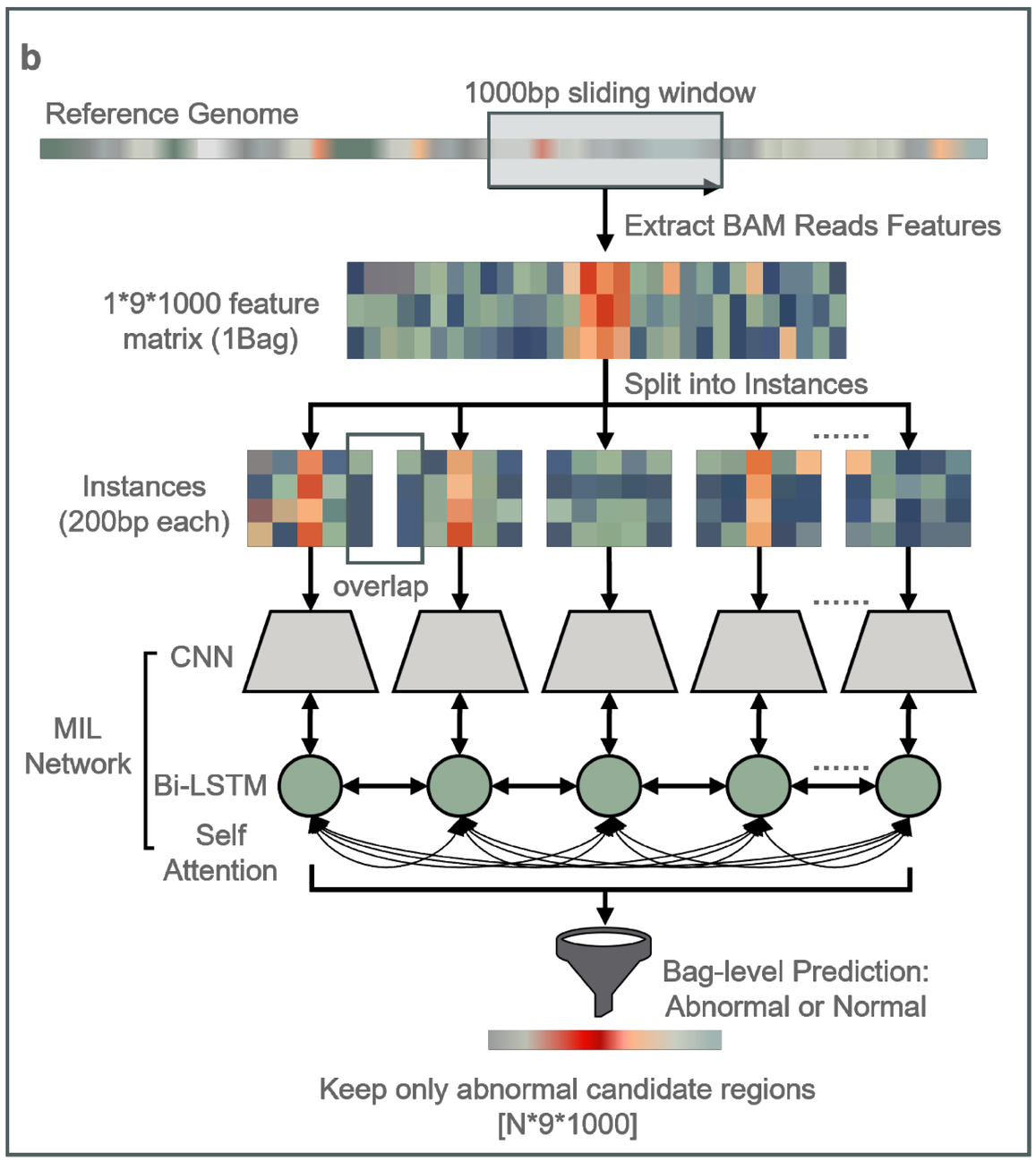
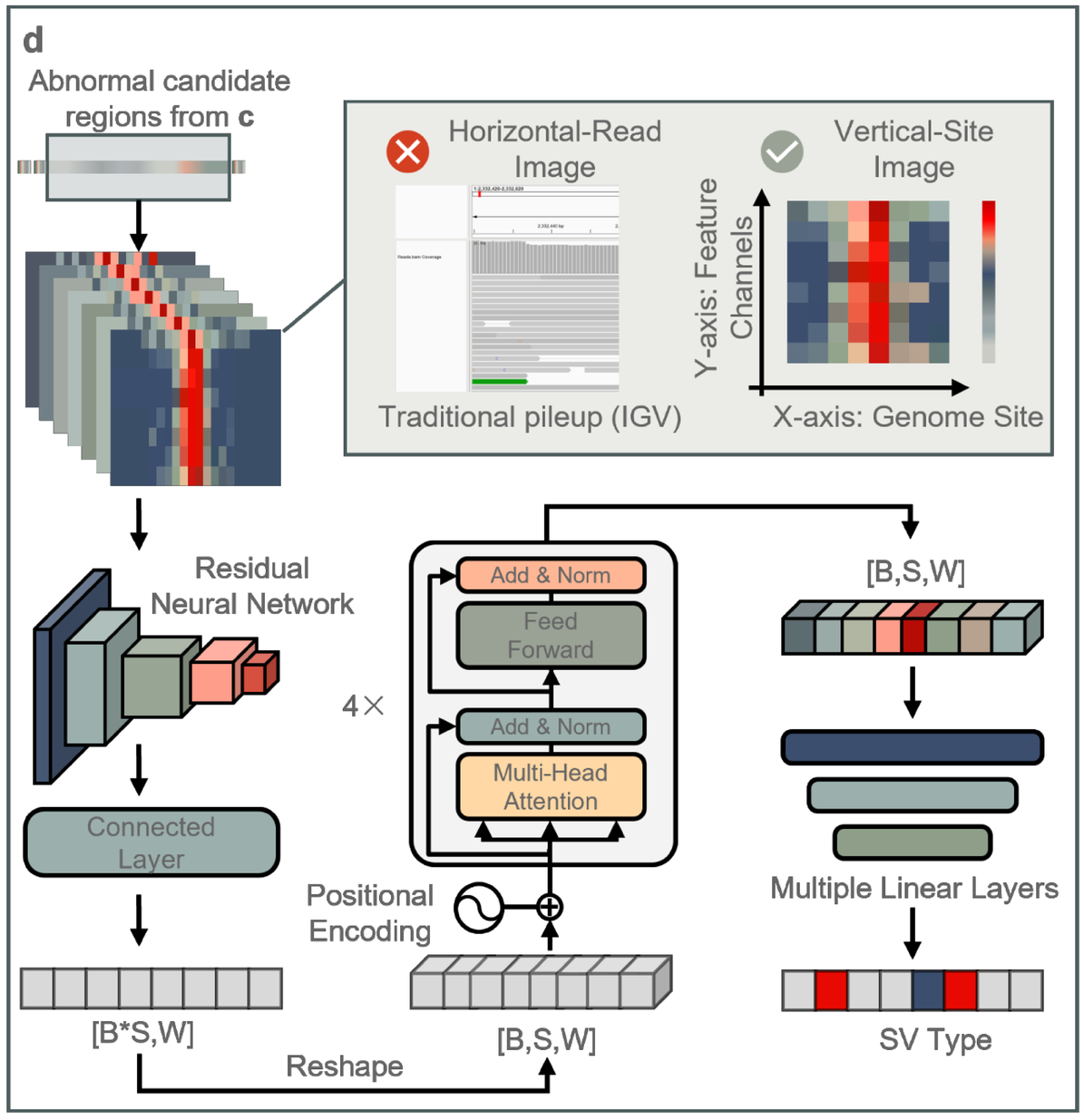
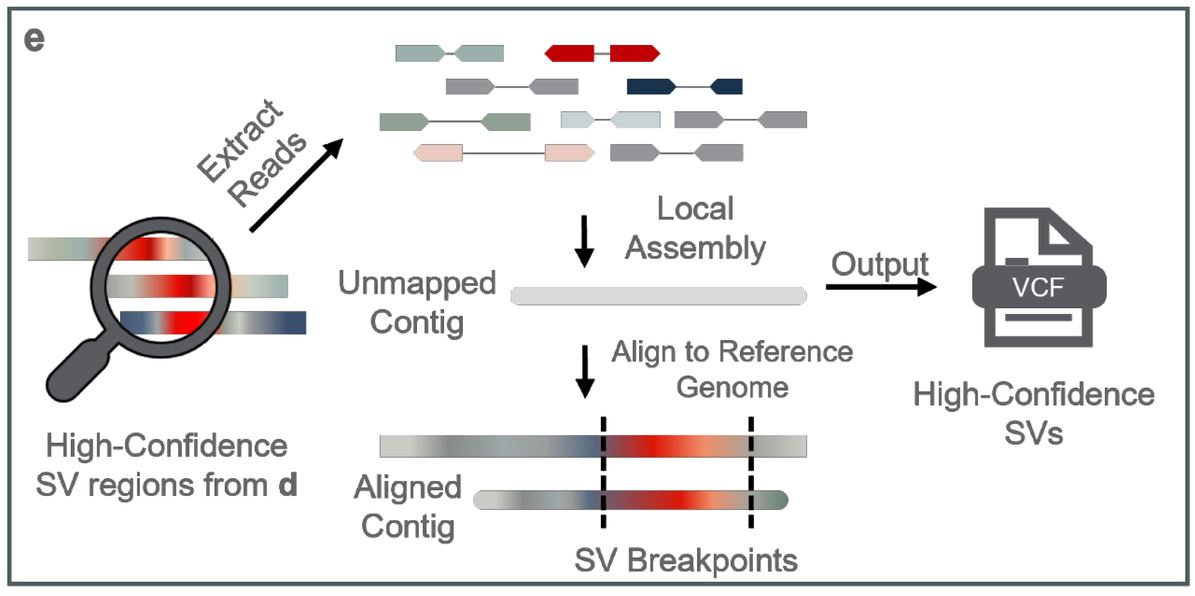
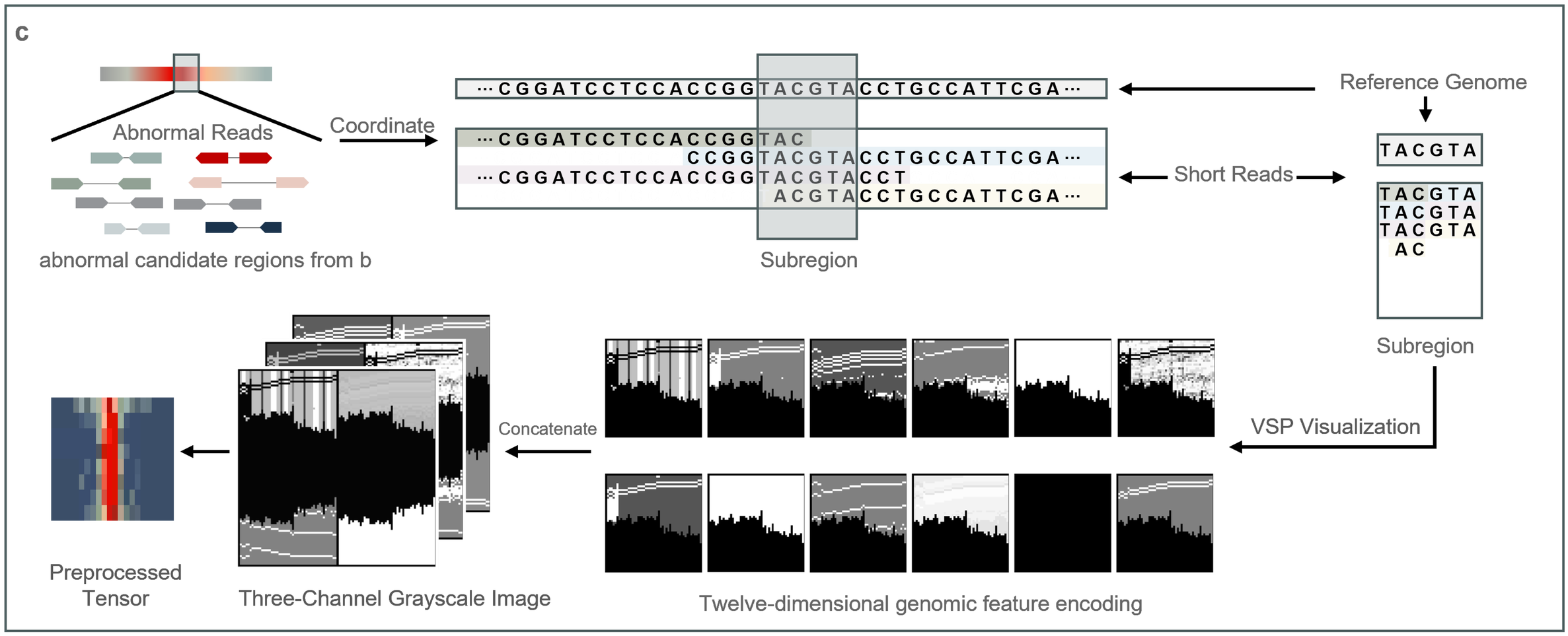

a
Reference Genome
Normal Regions
Abnormal Regions
Normal Read Pairs
Discordant Read Pairs And Split Reads
Coordinates


b
Reference Genome
1000bp sliding window
Extract BAM Reads Features
1*9*1000 feature matrix (1Bag)
Split into Instances
Instances (200bp each)
overlap
CNN
MIL Network
Bi-LSTM
Self Attention
Bag-level Prediction: Abnormal or Normal
Keep only abnormal candidate regions [N*9*1000]


d
Abnormal candidate regions from c
Horizontal-Read Image
Vertical-Site Image
Y-axis: Feature Channels
Traditional pileup (IGV)
X-axis: Genome Site
Residual Neural Network
Connected Layer
[B*S,W]
Reshape
4×
Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention
Positional Encoding
[B,S,W]
[B,S,W]
Multiple Linear Layers
SV Type


e
Extract Reads
High-Confidence SV regions from d
Local Assembly
Unmapped Contig
Output
VCF
High-Confidence SVs
Align to Reference Genome
Aligned Contig
SV Breakpoints


c
Abnormal Reads
abnormal candidate regions from b
Coordinate
···CGGATCCTCCACCGGTACGTACCTGCCATTCGA···
···CGGATCCTCCACCGGTAC
CCGGTACGTACCTGCCATTCGA···
···CGGATCCTCCACCGGTACGTACCT
ACGTACCTGCCATTCGA···
Subregion
Reference Genome
TACGTA
Short Reads
TACGTA
TACGTA
TACGTA
AC
Subregion
VSP Visualization
Concatenate
Preprocessed Tensor
Three-Channel Grayscale Image
Twelve-dimensional genomic feature encoding